

Project Approach (Roadmap)

Title: AI-Driven Retail Customer Behaviour Analytics and Business Feedback System

Domain

- Retail Analytics
- Customer Analytics
- Business Intelligence
- Data Analytics

Dataset

1. Online Retail II Dataset (UCI Machine Learning Repository)

<https://archive.ics.uci.edu/dataset/502/online+retail+ii>

2. Online Retail Dataset (UCI Machine Learning Repository)

<https://archive.ics.uci.edu/dataset/352/online+retail>

3. Online Retail Dataset (Kaggle)

<https://www.kaggle.com/datasets/luisrenterialezano/retail-sales-dataset>

Abstract

Title: Data-Driven Customer Behaviour Analytics for Retail Businesses

The rapid growth of the retail industry has resulted in the generation of massive volumes of customer transaction data, making it essential for businesses to understand customer purchasing behaviour and improve strategic decision-making. This project proposes a **Data-Driven Customer Behaviour Analytics System** that integrates data analytics, customer segmentation, anomaly detection, business intelligence, and machine learning techniques to transform raw retail data into meaningful business insights.

The project utilizes benchmark retail datasets from the **UCI Machine Learning Repository** and **Kaggle** containing customer transactions, product information, sales records, invoice details, and purchasing behaviour. The collected data undergoes preprocessing techniques including data cleaning, missing value handling, duplicate removal, outlier detection, encoding, feature scaling, and feature engineering.

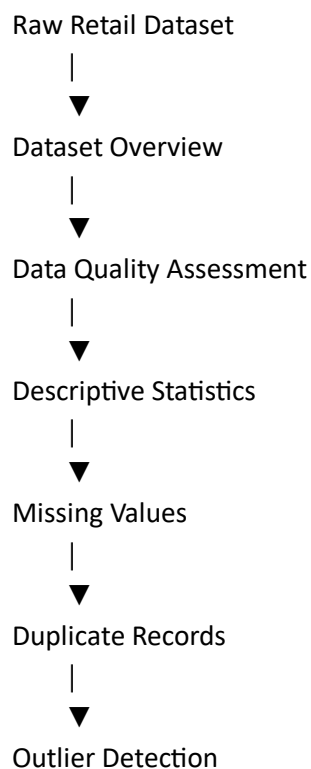
A comprehensive **Exploratory Data Analysis (EDA)** is performed using descriptive statistics, univariate, bivariate, and multivariate analysis, correlation analysis, customer segmentation, product performance analysis, customer purchase behaviour analysis, sales trend analysis, and regional sales analysis. To obtain deeper customer insights, **K-Means Clustering**, **DBSCAN**, **Isolation Forest**, and **Apriori Association Rule Mining** are implemented to identify customer segments, detect abnormal purchasing behaviour, and discover product purchasing patterns.

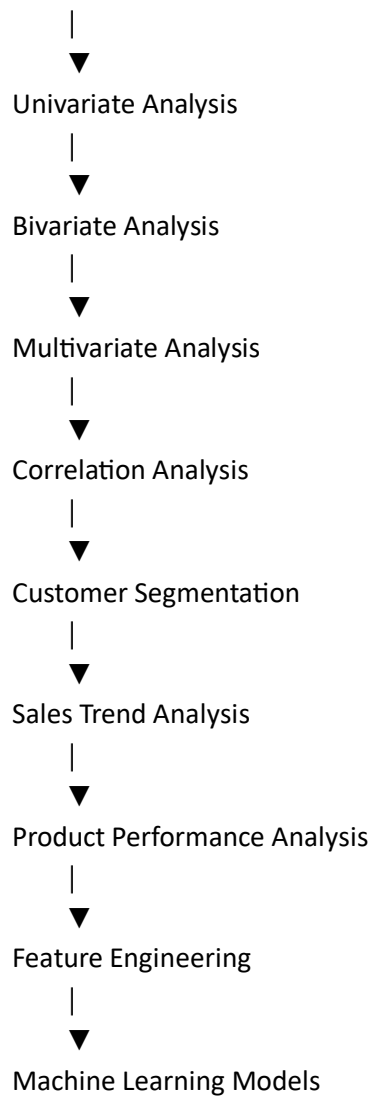
The analytical results are visualized using an interactive **Power BI Dashboard** consisting of customer segmentation, sales performance, product analysis, regional analysis, and key business performance indicators. Finally, the system generates an **Automated Customer Behaviour Verdict** by combining analytical findings, machine learning results, and business intelligence insights to provide actionable recommendations for improving customer retention, marketing strategies, inventory planning, and overall business performance.

The proposed system acts as an intelligent retail decision-support framework that enables organizations to make accurate, data-driven business decisions and improve operational efficiency and profitability.

Data Analytics

EDA Flow Diagram





Preprocessing

- Missing Value Handling
- Duplicate Removal
- Data Cleaning
- Encoding
- Feature Scaling / Normalization

Machine Learning Models

- K-Means Clustering
- DBSCAN
- Isolation Forest
- Apriori Association Rule Mining

Visualization

- Heatmap
- Boxplot

- Histogram
- Bar Chart
- Pie Chart
- Correlation Matrix
- Sales Trend Analysis
- Customer Segmentation Charts
- Dashboard

Business Intelligence

- Customer Segmentation Dashboard
- Sales Performance Dashboard
- Product Performance Dashboard
- Regional Sales Dashboard
- Customer Purchase Behaviour Dashboard
- Executive Business Dashboard

Comparative Analysis

Compare

- K-Means Clustering
- DBSCAN
- Isolation Forest
- Apriori Association Rule Mining

Metrics

- Silhouette Score
- Davies-Bouldin Index
- Calinski-Harabasz Index
- Support
- Confidence
- Lift
- Revenue Contribution
- Customer Retention Rate
- Average Order Value
- Repeat Purchase Rate

Automated Customer Behaviour Verdict

The system generates an overall business verdict based on customer segmentation, purchasing patterns, product performance, sales trends, anomaly detection, and association rule mining.

The verdict provides:

- Overall Business Health
- High Value Customer Segments
- Customer Purchase Insights
- Best Performing Products
- Weak Performing Regions
- Business Growth Opportunities
- Inventory Recommendations
- Marketing Recommendations
- Customer Retention Suggestions

Recommended Research Pipeline

Benchmark Retail Dataset



Data Cleaning



Exploratory Data Analysis



Customer Purchase Behaviour Analysis



Customer Segmentation



Sales Trend Analysis



Product Performance Analysis



Feature Engineering



Machine Learning Models



Business Intelligence Dashboard



Business Recommendation Engine

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Automated Customer Behaviour Verdict

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Research Conclusions